

A COLLEGE PROJECT BY YUVRAJ & HARWINDER

REAL-TIME CODE EDITOR

Collaborate. Code. Share instantly.

A browser-based room where two people edit the same file at the same time.

● LIVE SYNC

COLLABORATIVE ROOM

2 online ●

● Yuvraj

● Harwinder

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      cout << "Hello";
6  }
7  |
```

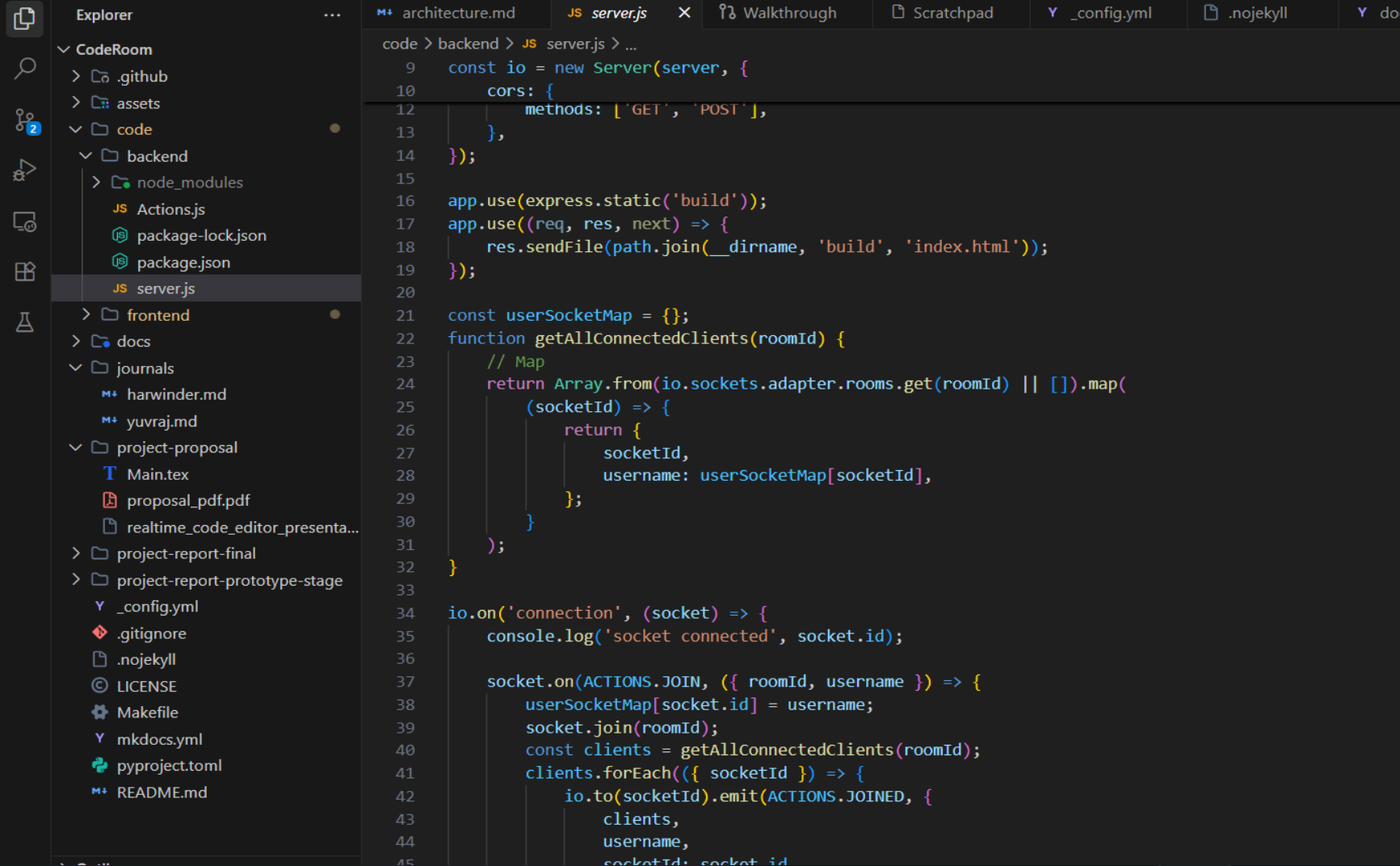
Harwinder is typing...

Connected



Tester

```
1 // The class name must exactly match the filename (e.g., Main.java)
2 public class Main {
3     // The entry point of any Java program
4     public static void main(String[] args) {
5         // Output text to the console
6         System.out.println("Hello, World!");
7     }
8 }
```



Copy ROOM ID

Leave

Paste invitation ROOM ID

b71fe067-4f36-44b7-b746-51027e10b9dc

USERNAME

Join

If you don't have an invite then create [new room](#)

When two people want to code together...

● Traditional workflow

- Share files manually
- Edit separately
- Send updated versions
- Merge changes later

● What goes wrong

- Version conflicts
- Repeated file sharing
- Delayed feedback
- No view of live changes

● In a classroom

- One student explains logic
- The other writes code
- Both need the same file
- Collaboration slows down

THE QUESTION

Can two people work on the same code, in the same room, at the same time?

One room. Multiple users. One editor.

Everyone joins with a Room ID and edits the same file — changes broadcast instantly.

ROOM #A7F92C

● Yuvraj

● Harwinder

● LIVE

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      cout << "Hello World";
6  }
```

Both cursors live in the same buffer — no file passing, no merge step.

● Create or join a room

A single Room ID is the whole invite.

● See connected users

Know who is in the room, live.

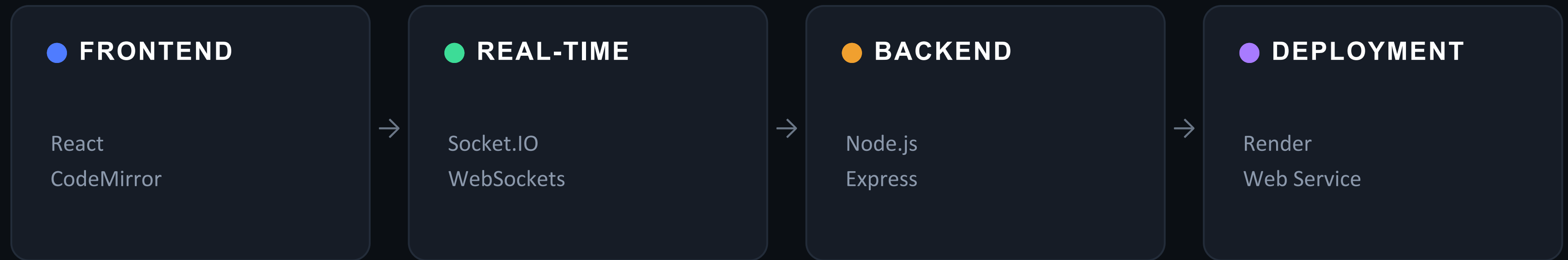
● Edit the same code

One shared buffer, not four copies.

● Broadcast in real time

Every keystroke reaches everyone.

Tech stack & data flow



REAL-TIME FLOW

User types → CodeMirror detects the change → Socket.IO emits the event → Node / Express broadcasts → **Everyone else updates**

Round trip is a single WebSocket hop — no polling, no page refresh.

Next step: C++ / Java / Python execution through a secure code-execution sandbox or API.

What changes for the user

0 file transfers

No sending versions back and forth. The room is the file.

1 shared buffer

Everyone edits the same code, so there is nothing to merge.

∞ live updates

Changes broadcast the moment they are typed.

Collaborate. Code. Share instantly.

Thank you — questions welcome

● Yuvraj

● Harwinder